# Programming Fundamentals

# Quiz-09

## Problem 01:

Write a function to print the following crown for height H:

```
*             *             *
**           ***           **
***         *****         ***
****       *******       ****
*****     *********     *****
****** *********** ******
***************************
***************************
***************************
***************************
***************************
```

Write a function to print the following crown for height H:
For H:5

```
*             *             *
 **          ***          **
  ***       *****       ***
   ****    *******    ****
    ******************
    ******************
     ****************
```

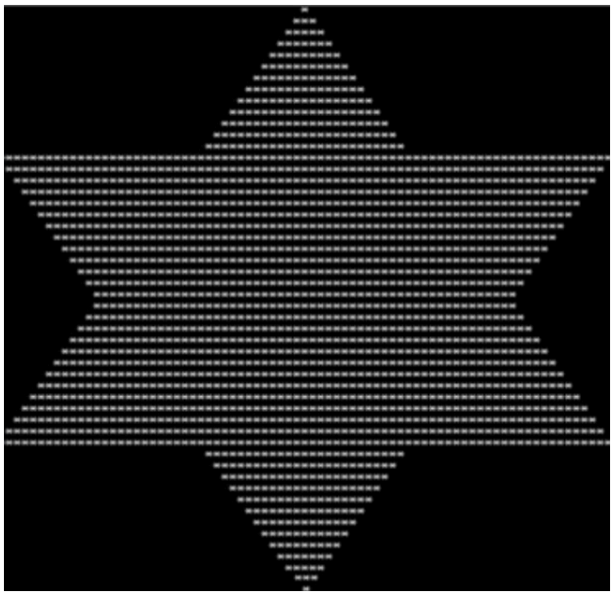
## Problem 02:

Observe the following pattern and write a code that prints the following pattern till given input.

```
Enter a character from A to Z: L
A
ABA
ABCBA
ABCDCBA
ABCDEDCBA
ABCDEFEDCBA
ABCDEFGFEDCBA
ABCDEFGHGFEDCBA
ABCDEFGHIHGFEDCBA
ABCDEFGHIJIHGFEDCBA
ABCDEFGHIJKJIHGFEDCBA
ABCDEFGHIJKLKJIHGFEDCBA
```

## Problem 03:

Observe the following pattern and write a code that prints the following pattern for height H.



## Problem 04:

Write helping functions that print the following patterns without nested loops  for height H:
H = 5

```
1
2  3
4  5  6
7  8  9  10
```

```
        1
      1 2 1
    1 2 3 2 1
  1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1
```

```
        1
      2 3 2
    3 4 5 4 3
  4 5 6 7 6 5 4
5 6 7 8 9 8 7 6 5
```

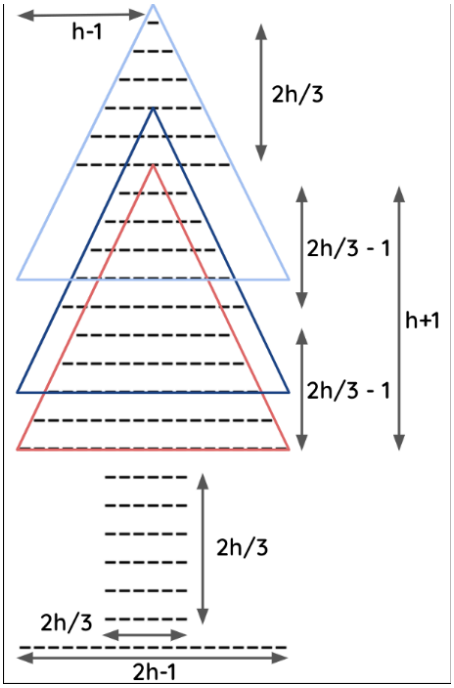
## Problem 05:

Write a function that prints the following shape for height H:(verify that the height will always be an odd number).
H = 5

```
    *1
   **12
  ***123
 ****1234
*****12345
13579ABCDE
 3579EFGH
  579HIJ
   79JK
    9K
```

## Problem 06:

Write a function that prints the following shape for height h:



```
O===O
 \O/
  |
 /\
```

Happy Coding... :)

Good luck